

KVM/QEMU Hypervisor Lab

Day 3 Lab Journal — SRV01 Confirmation, OS Downgrade Decision & SRV02 Deployment

Date: June 25, 2026 | Engineer: Demi (bravesoldier) | Host OS: Ubuntu 24.04.3 LTS

Placement: Navy QUORRA — SIWES Industrial Training

1. Day 3 Objective

Verify SRV01 (Ubuntu 24.04 Server) is operational, assess compatibility requirements for planned services, action a commandant-directed OS downgrade decision, deploy SRV02 on Ubuntu 20.04 LTS as the Docker and services host, and hand off Docker configuration to a colleague for continuation.

2. SRV01 Verification

2.1 Confirming SRV01 Is Running

Checked all VM states on the hypervisor:

```
virsh list --all
```

Both DC01 and SRV01 confirmed as running. SRV01 IP confirmed via libvirt:

```
virsh domifaddr SRV01
```

Output: IP assigned by DC01 DHCP as expected.

2.2 SSH Access to SRV01

SRV01 is on the isolated labnet. Direct SSH from a laptop on the 192.168.1.x network is not possible — the hypervisor acts as the gateway between both networks.

SSH into SRV01 was performed directly from the hypervisor terminal:

```
ssh username@srv01IP
```

Login successful. SRV01 confirmed operational on Ubuntu Server 24.04.3 LTS.

3. OS Downgrade Decision — Ubuntu 24.04 to 20.04

3.1 Compatibility Issue Identified

During service planning for SRV01, compatibility issues were identified between Ubuntu 24.04 and key services required for the lab — specifically Apache and associated applications. Ubuntu 24.04 ships with newer library versions that conflict with the target service stack.

3.2 Commandant Direction

The lab commandant reviewed the compatibility constraints and directed a downgrade to Ubuntu Server 20.04.6 LTS. Ubuntu 20.04 (Focal Fossa) provides better compatibility with the target Apache and Docker service stack and remains on Long Term Support until April 2025 (standard) / April 2030 (extended).

3.3 Decision — Keep SRV01, Deploy New SRV02

Rather than rebuilding SRV01 (which was freshly installed and may serve another purpose), the decision was made to deploy a separate VM — SRV02 — running Ubuntu 20.04.6 LTS as the dedicated Docker and services host. SRV01 remains available for future Ubuntu 24.04 compatible workloads.

4. Ubuntu 20.04 ISO Acquisition

4.1 Source — USB Flash Drive

Ubuntu 20.04.6 LTS Server ISO was available on a USB flash drive. Rather than downloading over the network, the ISO was transferred directly from the flash drive to the hypervisor's ISO storage on HDD1.

4.2 Flash Drive Detection

```
lsblk
```

Flash drive detected as sdc with partition sdc1.

4.3 Mount and Copy

```
sudo mkdir -p /mnt/usb
sudo mount /dev/sdc1 /mnt/usb
ls /mnt/usb
```

ISO confirmed: Ubuntu-20.04.6-live-server-amd64.iso

```
sudo cp /mnt/usb/Ubuntu-20.04.6-live-server-amd64.iso /mnt/data/isos/
sudo umount /mnt/usb
```

Transfer completed successfully. ISO verified in /mnt/data/isos/:

File	Size	Status
Ubuntu-20.04.6-live-server-amd64.iso	~1.4GB	✓ Ready
Ubuntu-24.04.3-live-server-amd64.iso	~3.1GB	✓ Ready
Windows_Server_2022.iso	~5.2GB	✓ Ready
virtio-win.iso	~500MB	✓ Ready

5. SRV02 Deployment — Ubuntu 20.04 LTS

5.1 VM Specifications

Parameter	Value
VM Name	SRV02
OS	Ubuntu Server 20.04.6 LTS (Focal Fossa)
vCPU	4
RAM	10GB (10240MB)
Disk	120GB qcow2
Network	labnet — 192.168.10.0/24 (DHCP)
Role	Docker host — Apache, web apps, databases, API services
Username	—

5.2 Disk Image Creation

```
sudo qemu-img create -f qcow2 /var/lib/libvirt/images/srv02.qcow2 120G
```

5.3 VM Launch

```
sudo env "PATH=$PATH" virt-install --name SRV02 --ram 10240 --vcpus=4  
--cpu host --disk  
path=/var/lib/libvirt/images/srv02.qcow2,format=qcow2,bus=virtio --cdrom  
/mnt/data/isos/Ubuntu-20.04.6-live-server-amd64.iso --os-variant  
ubuntu20.04 --network network=labnet,model=virtio --graphics vnc  
--noautoconsole --boot cdrom,hd
```

Output: Domain is still running. Installation may be in progress.

5.4 VNC Console Access

VNC port confirmed:

```
virsh vncdisplay SRV02
```

Output: 127.0.0.1:1 (port 5901)

SSH tunnel established from Windows laptop:

```
ssh -L 5912:127.0.0.1:5901 username@<hypervisor-ip>
```

Initial RealVNC connection failed — resolved by connecting to 127.0.0.1:5912 instead of localhost:5912.

5.5 Ubuntu 20.04 Installation — Steps

- Installer update — selected Update to new installer
- Language — English
- Keyboard — English (US)
- Installation type — Ubuntu Server (full, not minimized)
- Network — DHCP accepted, IP assigned from DC01 DHCP scope

- Proxy — left blank
- Mirror — default Ubuntu mirror
- Storage — Use entire disk with LVM
- Profile — name: — , server: srv02, username: —
- SSH — OpenSSH Server installed during setup
- Snaps — none selected
- Installation completed — VM rebooted automatically

5.6 SRV02 IP Confirmation

```
virsh domifaddr SRV02
```

SRV02 assigned IP: 192.168.10.120/24 via DC01 DHCP.

Note: If both SRV01 and SRV02 request DHCP leases simultaneously, they may receive different IPs. Static IPs should be configured on both servers in a future session to avoid conflicts.

5.7 SSH Access Confirmed

```
ssh username@srv01IP
```

Login successful from hypervisor terminal. SRV02 confirmed operational on Ubuntu 20.04.6 LTS.

6. Docker Configuration Handoff

Docker installation and service configuration on SRV02 was handed off to a colleague (Alex) for continuation. The following services are planned for deployment:

- Docker Engine and Docker Compose
- Apache/Nginx web server
- PostgreSQL or MySQL database
- Git, Python, Node.js development tools
- Internal API services

Progress will be assessed and documented on resumption. Day 4 will pick up from wherever the colleague left off.

7. Errors & Resolutions Summary

Error	Root Cause	Resolution
RealVNC connection failed on localhost:5912	RealVNC on Windows handled localhost differently than expected	Connected using 127.0.0.1:5912 instead
SRV01 not directly SSH accessible from laptop	labnet is an isolated NAT network — laptop is on 192.168.1.x	SSH into SRV01 via hypervisor terminal as stepping stone

Ubuntu 24.04 incompatible with target services	Apache and associated apps conflicted with Ubuntu 24.04 libraries	Commandant directed downgrade — new VM SRV02 deployed on Ubuntu 20.04
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8. End of Day 3 Status

Milestone	Status
SRV01 (Ubuntu 24.04) confirmed operational	✓ Complete
SRV01 SSH access verified	✓ Complete
OS downgrade decision documented and actioned	✓ Complete
Ubuntu 20.04 ISO transferred from USB to hypervisor	✓ Complete
SRV02 VM created (120GB, 10GB RAM, 4 vCPU)	✓ Complete
Ubuntu 20.04.6 LTS installed on SRV02	✓ Complete
SRV02 SSH access verified	✓ Complete
Docker Engine installation on SRV02	🕒 Day 4 (colleague handoff)
Apache/Nginx configuration	🕒 Day 4
Database setup (PostgreSQL/MySQL)	🕒 Day 4
Static IP configuration for SRV01 and SRV02	🕒 Day 4

Day 4 — Next Steps

- Assess Docker progress made by colleague on SRV02
- Complete Docker Engine and Docker Compose installation if not done
- Deploy Apache/Nginx container
- Deploy database container (PostgreSQL or MySQL)
- Set static IPs on SRV01 and SRV02 Test connectivity between DC01, SRV01, and SRV02
- Begin domain join process for Linux VMs